iteratively in terms of its children – for practical reasons, though, to improve the running time of the search methods, in practice we store for each rectangle its bounding box.

Since the rectangles can only be parallel to the cartesian axes, they are defined by two of their vertices: two tuples with k coordinates, one tuple for the minimum values of each coordinate and one for the maximum value of each coordinate.

Notice how, differently from k-d trees, an R-tree could handle a non-zero-measure object, by simply considering its bounding boxes as special cases of rectangles, as illustrated in figure 10.5.

10.2.3 Inserting Points in an R-tree

Now, of course, you might legitimately wonder how do you get from a raw dataset to the R-tree in figure 10.5; after all, we have just presented it and asked you to take it as a given, just as it is.

Insertion for R-trees is similar to B-trees, and has many steps in common with SS-trees, so we won't duplicate the learning effort with a detailed description here.

At a high level, to insert a new point you will need the following steps:

1. Find the leave that should host the new point P. There are three possible cases: